Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Equation of the directrix of the parabola $y^2 + 4y + 4x + 2 = 0$ is

(A) $x = -1$

(B) $x = 1$

(C) $x = -3/2$

(D) $x = 3/2$

Q.2

If the line $x - 1 = 0$ is the directrix of the parabola $y^2 - kx + 8 = 0, k \neq 0$ and the parabola intersects the circle $x^2 + y^2 = 4$ in two real distinct points, then the value of k is

(A) -4

(B) -8

(C) 4

(D) none of these

Q.3

An equation of the parabola whose focus is $(-3, 0)$ and the directrix is $x + 5 = 0$ is:

(A) $y^2 = 4(x + 5)$

(B) $y^2 = 4(x + 4)$

(C) $y^2 = 4(x + 3)$

(D) $y^2 = 4(x - 3)$

Q.4

PQ is a double ordinate of a parabola $y^2 = 4ax$. The locus of its points of trisection is another parabola length of whose latus rectum is k times the length of the latus rectum of the given parabola, the value of k is

(A) $1/9$

(B) $1/3$

(C) $2/3$

(D) none of these

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Q.5

$y^2 = 16x$ is a parabola and $x^2 + y^2 = 16$ is a circle. Then

(A) circle passes through the vertex of the parabola

(B) circle touches the parabola at the vertex

(C) circle passes through the focus of the parabola

(D) circle lies inside the parabola.

Q.6

The focus of the parabola $4y^2 + 12x - 20y + 67 = 0$ is

(A) $(-7/2, 5/2)$

(B) $(-3/4, 5/2)$

(C) $(-17/4, 5/2)$

(D) $(5/2, -3/4)$

Q.7

Consider the parabolas $S_1 \equiv y^2 - 4ax = 0$ and
$S_2 \equiv y^2 - 4bx = 0. S_2$ will contain S_1, if

(A) $a > b > 0$

(B) $b > a > 0$

(C) $a > 0, b < 0$ but $|b| > a$

(D) $a < 0, b > 0$ but $b > |a|$

Q.8

What is the area of the triangle formed by the lines joining the vertex of the parabola $x^2 = 12y$ to the ends of the latus rectum?

(A) 9 square units

(B) 12 square units

(C) 14 square units

(D) 18 square units

Q.9

The locus of the vertices of the family of parabolas $y = \frac{a^3x^2}{3} + \frac{a^3x}{2} - 2a$ is

(A) $xy = \frac{64}{105}$

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(B) $xy = \frac{105}{64}$

(C) $xy = \frac{3}{4}$

(D) $xy = \frac{35}{16}$

Q.10

A parabola has the origin as its focus and the line $x = 2$ as the directrix. Then the vertex of the parabola is at

(A) $(1, 0)$

(B) $(0, 1)$

(C) $(2, 0)$

(D) $(0, 2)$

Q.11

If $a \neq 0$ and the line $2bx + 3cy + 4d = 0$ passes through the points of intersection of the parabolas $y^2 = 4ax$ and $x^2 = 4ay$, then

(A) $d^2 + (2b + 3c)^2 = 0$

(B) $d^2 + (3b - 2c)^2 = 0$

(C) $d^2 + (2b - 3c)^2 = 0$

(D) $d^2 + (3b + 2c)^2 = 0$

Q.12

A line with slope m passing through focus cuts the parabola $y^2 = 4x$ at two distinct points, then

(A) $m \in R$

(B) $m \in R^+$

(C) $0 \leq m \leq 1$

(D) none of these

Q.13

The coordinates of a point on the parabola $y^2 = 8x$ whose focal distance is 4 , are

(A) $\left(\frac{1}{2}, \pm 1\right)$

(B) $(1, \pm 2\sqrt{2})$

(C) $(2, \pm 4)$

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(D) None of these

Q.14

The length of a focal chord of the parabola $y^2 = 4ax$ making an angle θ with the axis of the parabola is

(A) $4a \operatorname{cosec}^2 \theta$

(B) $4a \sec^2 \theta$

(C) $a \operatorname{cosec}^2 \theta$

(D) None of these

Q.15

An equilateral triangle is inscribed in the parabola $y^2 = 4ax$ whose one of the vertex is at the vertex of the parabola. The length of its side is

(A) $2a\sqrt{3}$

(B) $4a\sqrt{3}$

(C) $60\sqrt{3}$

(D) $8a\sqrt{3}$

Q.16

The focus of the parabola $y^2 - x - 2y + 2 = 0$ is

(A) $(1/4, 0)$

(B) $(1, 2)$

(C) $(3/4, 1)$

(D) $(5/4, 1)$

Q.17

The point $(a, 2a)$ is an interior point of region bounded by the parabola $x^2 = 16\mathrm{y}$ and the double ordinate through focus then

(A) $a < 4$

(B) $0 < a < 4$

(C) $0 < a < 2$

(D) $a > 4$

Q.18

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The conic represented by the equation $\sqrt{ax} + \sqrt{by} = 1$ is

(A) ellipse

(B) hyperbola

(C) parabola

(D) None of these

Q.19

Curve $x^2 + (1+m)y^2 - 4mx - 5 = 0$ always passes through

(A) one fixed point

(B) two fixed points

(C) three fixed points

(D) four fixed points

Q.20

Equation of the parabola if its vertex and focus are $(-2, 3)$ and $(1, -1)$ respectively, is

(A) $16x^2 + 24xy + 9y^2 - 308x + 394y - 1799 = 0$

(B) $16x^2 + 24xy + 9y^2 - 308x - 394y - 1799 = 0$

(C) $16x^2 - 24xy + 9y^2 - 308x + 394y - 1799 = 0$

(D) None of these

Answers & Solutions

Q.1 Answer:

$x = 3/2$

Solution:

Given equation can be written as $(y+2)^2 = -4x + 2 = -4(x - 1/2)$
which is of the form $Y^2 = 4aX$ **where** $Y = y + 2, X = x - 1/2, a = -1$
The directrix of the parabola $Y^2 = 4aX$ **is** $X = -a \Rightarrow \quad x - 1/2 = -(-1) \quad \Rightarrow \quad x = 3/2$
is the equation of the directrix of the given parabola

Q.2 Answer:

-8

Solution:

The equation of the parabola can be written as $y^2 = k(x - 8/k)$ **which is of the form** $Y^2 = 4AX$

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where $Y = y, X = x - 8/k$ and $A = k/4$

Equation of the directrix is

$X = -A \Rightarrow x - 8/k = -k/4$

which represents the given line $x - 1 = 0$ if $\frac{8}{k} - \frac{k}{4} = 1$

$\Rightarrow k^2 + 4k - 32 = 0 \Rightarrow k = -8$ or 4

For $k = 4$, the parabola is $y^2 = 4(x - 2)$ whose vertex is $(2, 0)$ and touches the circle $x^2 + y^2 = 4$ at the vertex. Therefore $k \neq 4$.

Q.3 Answer:

$y^2 = 4(x + 4)$

Solution:

We have $\sqrt{(x+3)^2 + y^2} = |x + 5|$

$\Rightarrow y^2 = (x + 5)^2 - (x + 3)^2 = 4(x + 4)$

Q.4 Answer:

$1/9$

Solution:

Let PQ be a double ordinate of the parabola $y^2 = 4ax$ meeting the axis at L, R and S be the points of trisection of PQ. So that $PR = RS = SQ$. (Fig.)

If the coordinates of R are (h, k), the coordinates of S are $(h, -k)$

the coordinates of P are $(h, 3k)$

Since P lies on the parabola $y^2 = 4ax$

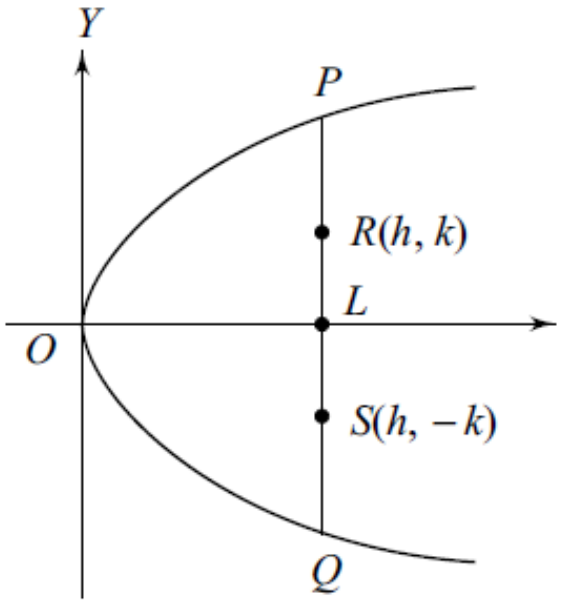
$(3k)^2 = 4ah \quad \Rightarrow \quad 9k^2 = 4ah$

The locus of $R(h, k)$ is therefore

$9y^2 = 4ax$ or $y^2 = (4/9)ax$

which is a parabola whose length of the latus rectum $= (4a/9)$

$= (1/9)$ (length of latus rectum of the given parabola)



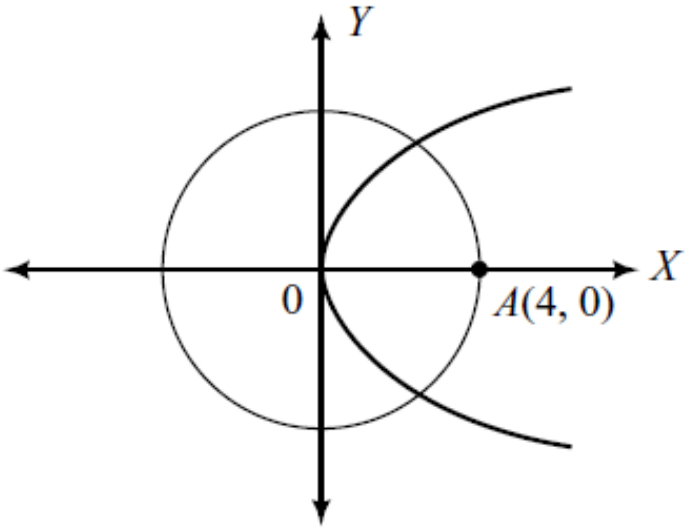
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Q.5 Answer:

circle passes through the focus of the parabola

Solution:

Centre of the circle is the origin, the vertex of the parabola and the focus of the parabola is $A(4,0)$. The circle passes through A.



Q.6 Answer:

$(-17/4, 5/2)$

Solution:

The given equation of the parabola can be written as $y^2 - 5y = -3x - 67/4 \Rightarrow (y - 5/2)^2 = -3(x + 7/2)$

$\Rightarrow \quad Y^2 = 4aX$ **where** $Y = y - 5/2, X = x + 7/2$ **and** $a = -3/4$

The focus of $Y^2 = 4aX$ **is** $(X, Y) = (a, 0) = (-3/4, 0) \Rightarrow \quad x + 7/2 = -3/4, y - 5/2 = 0 \Rightarrow x = -17/4, y = 5/2$

Therefore, required focus is $(-17/4, 5/2)$

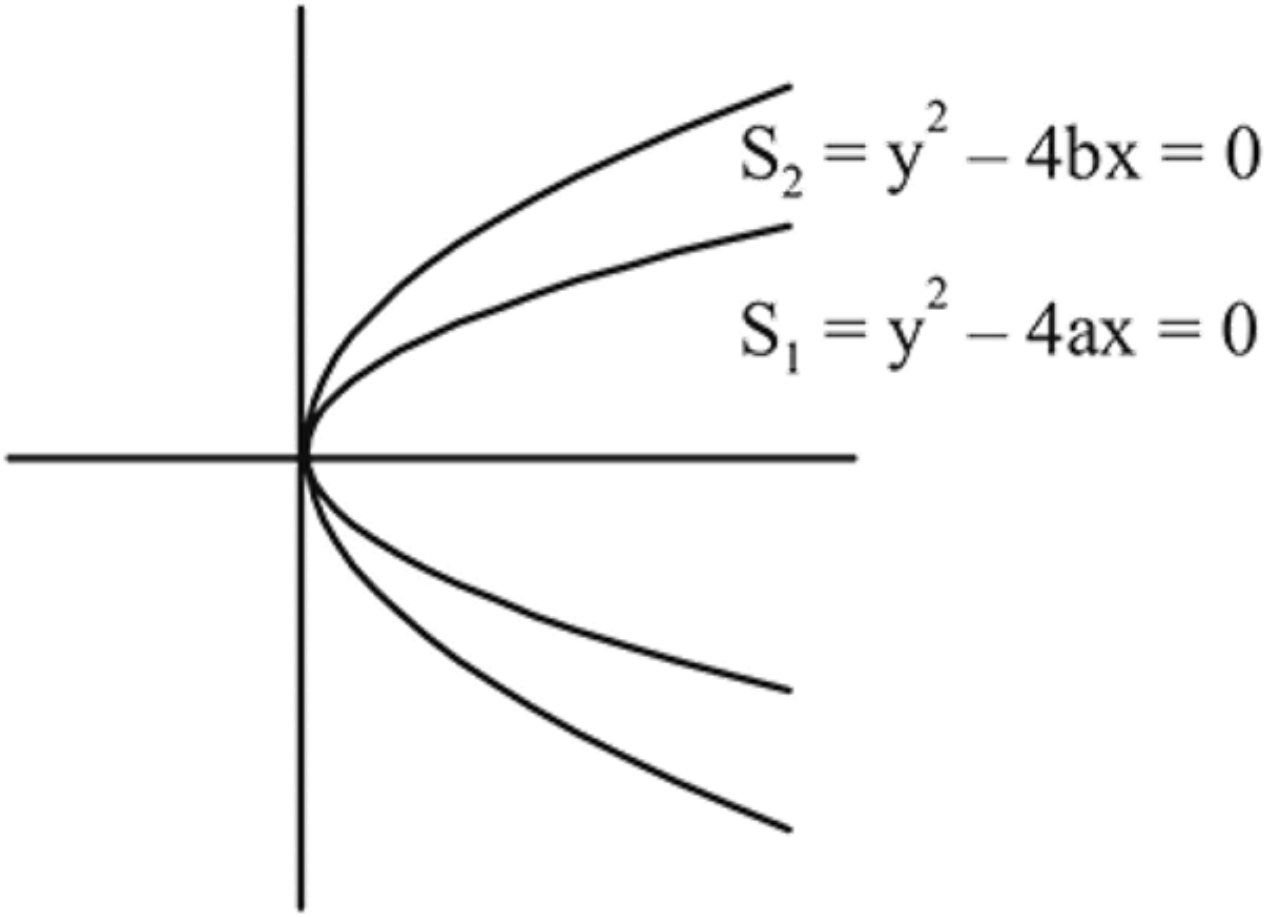
Q.7 Answer:

$b > a > 0$

Solution:

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If a and $b > 0$, then graphic representation would be as follows:



S_2 will contain S_1 if latusrectum of S_2 > latusrectum of $S_1 \Rightarrow 4b > 4a$

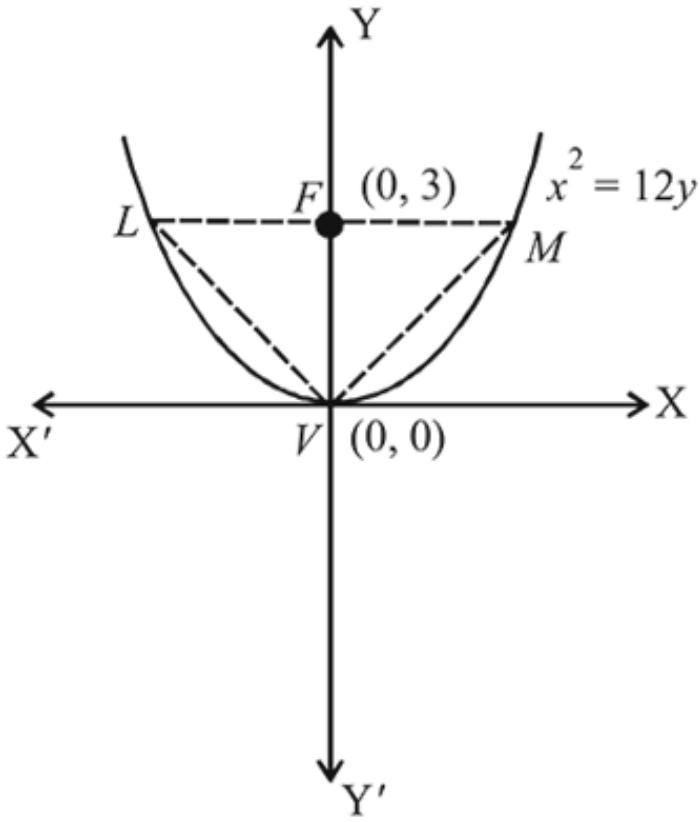
$\therefore b > a > 0$

Q.8 Answer:

18 square units

Solution:

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Given parabola is $x^2 = 12y$ which is of the form $x^2 = 4ay$

$\Rightarrow 4a = 12 \Rightarrow a = 3$

Now, LM is the latus rectum whose length $= 4a = 4 \times 3 = 12$

So, area of $\Delta LMV = \frac{1}{2} \times LM \times VF.$

$= \left(\frac{1}{2} \times 12 \times 3\right)$ **sq. unit**

$= 18$ **square unit**

Q.9 Answer:

$xy = \frac{105}{64}$

Solution:

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Given family of parabolas

$$y = \frac{a^3x^2}{3} + \frac{a^2x}{2} - 2a$$

$$\Rightarrow \frac{y}{a^3/3} = x^2 + \frac{a^2 \cdot 3}{2 \cdot a^3}x - \frac{2a}{a^3/3}$$

$$\Rightarrow \frac{3y}{a^3} + \frac{6a}{a^3} = x^2 + 2\left(\frac{3}{4a}\right)x + \frac{9}{16a^2} - \frac{9}{16a^2}$$

$$\Rightarrow \frac{3y}{a^3} + \frac{6a}{a^3} + \frac{9}{16a^2} = \left(x + \frac{3}{4a}\right)^2$$

$$\Rightarrow \frac{3y}{a^3} + \frac{6}{a^2} + \frac{9}{16a^2} = \left(x + \frac{3}{4a}\right)^2$$

$$\Rightarrow \left(x + \frac{3}{4a}\right)^2 = \frac{3y}{a^3} + \frac{105}{16a^2}$$

$$\Rightarrow \left(x + \frac{3}{4a}\right)^2 = \frac{3}{a^3}\left(y + \frac{35}{16}a\right) \quad \ldots(i)$$

Let $x + \frac{3}{4a} = X$ **and** $y + \frac{35}{16}a = Y$

$\therefore$ equation (i) reduces to

$X^2 = \frac{3}{a^3}Y \Rightarrow$ **vertex**

$X = 0, Y = 0$

$\Rightarrow x + \frac{3}{4a} = 0 \Rightarrow x = -\frac{3}{4a}$

$y + \frac{35}{16}a = 0 \Rightarrow y = -\frac{35}{16}a$

$\therefore xy = -\frac{3}{4a} \times \left(-\frac{35}{16}a\right) = \frac{105}{64}$

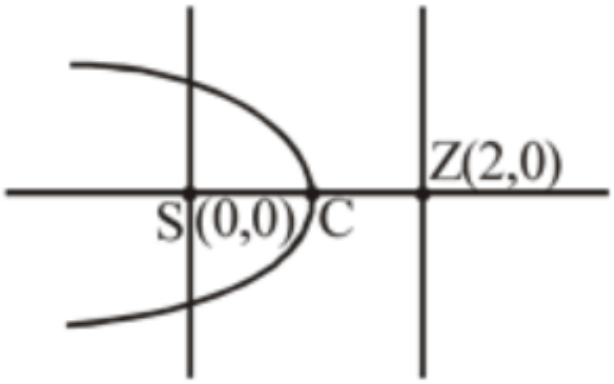
$\therefore$ required locus is : $xy = \frac{105}{64}$

Q.10 Answer:

$(1, 0)$

Solution:

Vertex is mid-point of S **and** Z $\Rightarrow$ **vertex**



$= (1, 0)$

Q.11 Answer:

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$d^2 + (2b + 3c)^2 = 0$

Solution:

The parabolas $y^2 = 4\alpha x$ and $x^2 = 4ay$ meet at $(0, 0)$ and $(4a, 4a)$.
These points lie on $2bx + 3cy + 4d = 0$.
Hence, $d = 0$ and $2b + 3c = 0$

Q.12 Answer:

none of these

Solution:

Focus of the parabola $= (1, 0)$
Equation of line with slope m and passing through focus
$(y - 0) = m(x - 1) \Rightarrow y = mx - m$
Now it cuts the parabola at two distinct points so,
$m^2(x - 1)^2 = 4x \Rightarrow m^2\left(x^2 + 1 - 2x\right) = 4x$
$\Rightarrow m^2x^2 - 2x\left(m^2 + 2\right) + m^2 = 0$ **...(1)**
For two distinct roots, discriminant should be greater then zero.
i.e. $\left[2\left(m^2 + 2\right)\right]^2 - 4m^4 > 0 \Rightarrow \left(m^2 + 1\right) > 0$ **which is true ";** $m \in R$
But for (1) to be a quadratic equation $m \neq 0$.
So, $m \in R - \{0\}$

Q.13 Answer:

$(2, \pm 4)$

Solution:

If the coordinates of a point on the parabola $y^2 = 4ax$ are $P(x, y)$, then its focal distance is
$SP = x +$ **a. Here** $a = 2$ **and** $\mathrm{SP} = 4$
$\therefore 4 = x + 2 \Rightarrow x = 2 \Rightarrow y^2 = 8 \times 2 \Rightarrow y = \pm 4$
Thus, the coordinates of required point are $(2, \pm 4)$

Q.14 Answer:

$4a\,\mathrm{cosec}^2\,\theta$

Solution:

Length of the focal chord is
$a\sqrt{\left(t^2 - \frac{1}{t^2}\right)^2 + 4\left(t + \frac{1}{t}\right)^2} = a\left(t + \frac{1}{t}\right)^2$
Now, $\tan\theta = \frac{2\left(t + \frac{1}{t}\right)}{t^2 - \frac{1}{t^2}} = \frac{2}{t - \frac{1}{t}} \Rightarrow t - \frac{1}{t} = 2\cot\theta$
$\left(t + \frac{1}{t}\right)^2 = \left(t - \frac{1}{t}\right)^2 + 4 = 4\cot^2\theta + 4$
$= 4\,\mathrm{cosec}^2\,\theta$
$\therefore$ **Length of the focal chord in terms of θ is** $4a\,\mathrm{cosec}^2\,\theta$

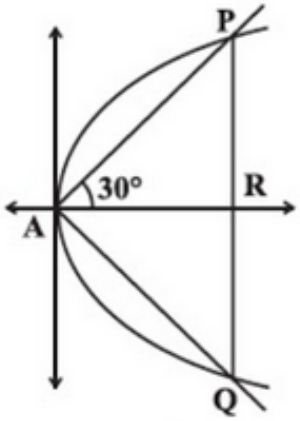
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Q.15 Answer:

$8a\sqrt{3}$

Solution:

As shown in the figure APQ denotes the equilateral triangle with its equal sides of length l (say).



$\mathrm{AP} = l$ so $\mathrm{AR} = l\cos 30^\circ$

$= l\dfrac{\sqrt{3}}{2}$

$\mathrm{PR} = l\sin 30^\circ = \dfrac{l}{2}$

Thus $(l\sqrt{3}/2, l/2)$ **are the coordinates of the point** P **lying on the parabola** $y^2 = 4ax$

Therefore,

$\frac{l^2}{4} = 4a\left(\frac{l\sqrt{3}}{2}\right) \Rightarrow l = 8a\sqrt{3}$

Thus, $8a\sqrt{3}$ **is the required length of the side of the equilateral triangle inscribed in the parabola** $y^2 = 4ax$

Q.16 Answer:

$(5/4, 1)$

Solution:

$y^2 - 2y = x - 2$ **or** $(y-1)^2 = 1.(x-1)$

$4a = 1$, **focus is** $X = A, Y = 0$

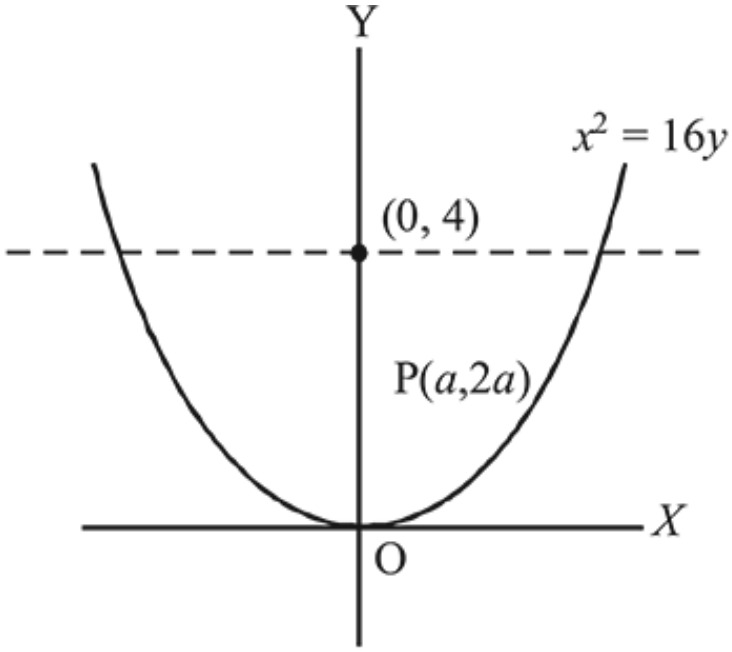
$x - 1 = \frac{1}{4}, y - 1 = 0 \therefore (5/4, 1)$

Q.17 Answer:

$0 < a < 2$

Solution:

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Parabola $x^2 = 16y$, axis is the y -axis, focus $= (0, 4)$ It is clear if point $P(a, 2a)$ lies interior region then ordinate $\in (0, 4) \Rightarrow 0 < 2a < 4 \Rightarrow 0 < a < 2$

Q.18 Answer:

parabola

Solution:

The given conic is $\sqrt{ax} + \sqrt{by} = 1$

Squaring both sides, $ax + by + 2\sqrt{abxy} = 1$

or $ax + by - 1 = -2\sqrt{abxy}$

Squaring again, $(ax + by - 1)^2 = 4abxy$

or $\mathrm{a^2x^2 - 2abxy + b^2y^2 - 2ax - 2by + 1 = 0}$ ...(1)

Comparing the equation (1) with the equation $\mathrm{Ax^2 + 2Hxy + By^2 + 2Gx + 2Fy + C = 0}$

$\therefore \mathrm{A = a^2, H = -ab, B = b^2, G = -a, F = -b, C = 1}$

Then, $\Delta = \mathrm{ABC + 2FGH - AF^2 - BG^2 - CH^2}$

$= a^2b^2 - 2a^2b^2 - a^2b^2 - a^2b^2 - a^2b^2 = -4a^2b^2 \neq 0$

and $\mathrm{H^2 = a^2\ b^2 = AB}$

So we have $\Delta \neq 0$ and $H^2 - AB = 0$. Hence the given equation represent a parabola.

Q.19 Answer:

two fixed points

Solution:

The given curve is

$x^2 + y^2 - 5 + m\left(y^2 - 4x\right) = 0$

It passes though the point of intersection of $x^2 + y^2 = 5$ and $y^2 = 4x$

$\therefore x = 1, -5$

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Here $x = 1 \Rightarrow y = \pm 2$ and
$x = -5 \Rightarrow y^2 = -20$ not possible.

Q.20 Answer:

$16x^2 + 24xy + 9y^2 - 308x + 394y - 1799 = 0$

Solution:

Distance between focus and vertex $= \sqrt{(1+2)^2 + (-1-3)^2} = 5$

Let point of intersection of directrix and axis of parabola be (x_1, x_1)**, then** $\frac{x_1+1}{2} = -2 \Rightarrow x_1 = -5$

and $\frac{y_1-1}{2} = 3, \Rightarrow y_1 = 7$

$\therefore$ **Equation of the directrix is** $y - 7 = \frac{3}{4}(x+5) \quad \Rightarrow 3x - 4y + 43 = 0$

$\therefore$ **Equation of required parabola is**

$\left(\frac{3x-4y+43}{5}\right)^2 = (x-1)^2 + (y+1)^2 \Rightarrow 16x^2 + 24xy + 9y^2 - 308x + 394y - 1799 = 0$

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